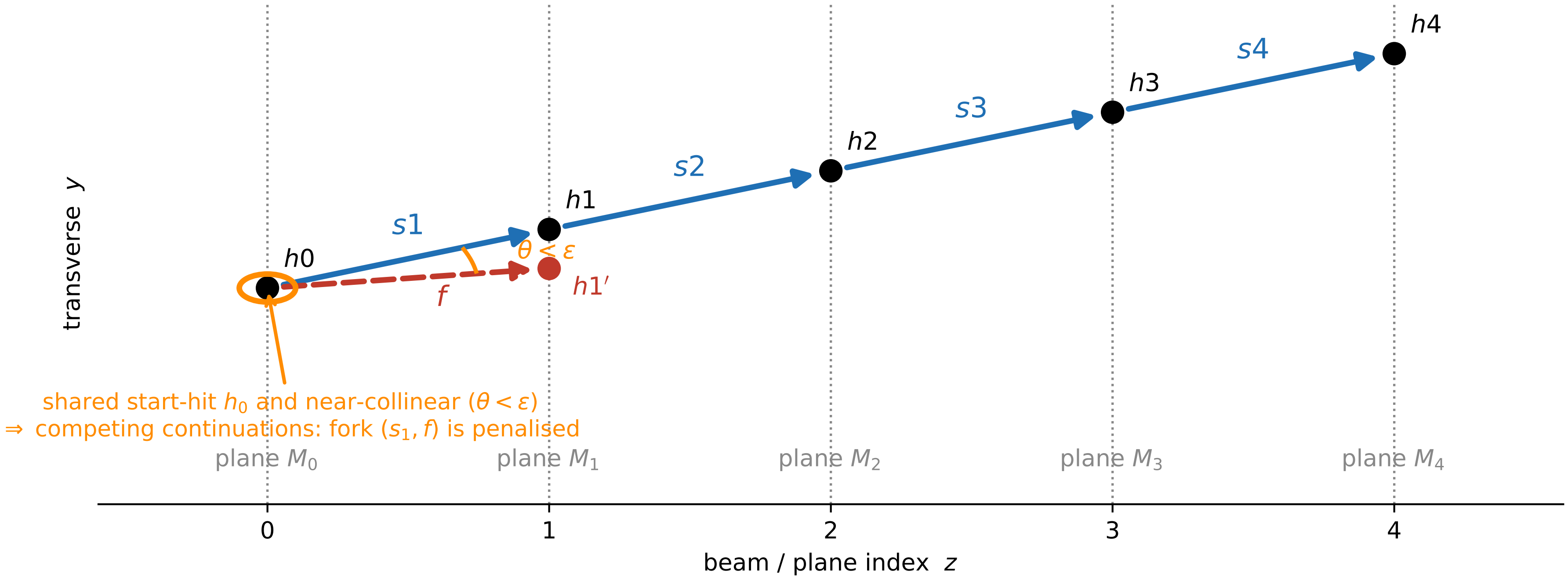
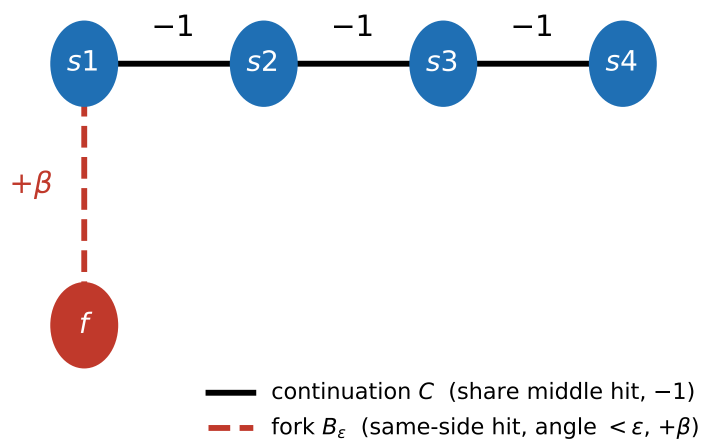
Bifurcation term — from labelled picture to Hamiltonian matrix

(a) Detector schematic: a true 4-segment chain $s_1 - s_2 - s_3 - s_4$ and a false fork f — same start-hit, near-collinear



(b) The two graphs the picture defines



(c) $A' = (\gamma+\delta)\, I - C + \beta B_\varepsilon$ (here $\gamma+\delta = 4$)

	$s1$	$s2$	$s3$	$s4$	f
$s1$	$\gamma+\delta$	-1	0	0	$+\beta$
$s2$	-1	$\gamma+\delta$	-1	0	0
$s3$	0	-1	$\gamma+\delta$	-1	0
$s4$	0	0	-1	$\gamma+\delta$	0
f	$+\beta$	0	0	0	$\gamma+\delta$

$\mathbf{b}' = \delta\mathbf{1}$ (off-diagonal form); full form: diag $\to \gamma+\delta+2\beta$, $\mathbf{b}'' = (\delta+\beta)\mathbf{1}$